Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Build and Deploy Agentic AI Apps with CrewAI, Autogen, ADK and Streamlit

TGS Ref No: TGS-2025059028

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 2

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	11 August 2026	Initial release — 4-day (32-hour) lesson plan for TGS-2025059028, aligned to the 20 hands-on activities across LU1-LU3 and to TSC Generative AI Application Development and Deployment (ICT-INT-0047-1.1).	Dr. Alfred Ang
2.0	11 August 2026	Revision after courseware QA — Day 3/4 schedule rebalanced so all 20 activities are scheduled (Activities 19 and 20 added to Day 4); duplicate slide citation removed; assessment time shown separately from the 32 instructional hours.	Dr. Alfred Ang

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Course Information

Course Title	Build and Deploy Agentic AI Apps with CrewAI, Autogen, ADK and Streamlit
WSQ Course Reference	TGS-2025059028
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Skills Framework TSC	Generative AI Application Development and Deployment (ICT-INT-0047-1.1), Level 1
Duration	4 days · 8 scheduled hours per day (32 hours), of which 30 hours are facilitation and the final 2 hours on Day 4 are the assessment
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Mode	Instructor-led, hands-on development labs in Python, Streamlit and the agent frameworks
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Design and implement multi-agent integration workflows using Agentic AI frameworks — CrewAI, AutoGen and the OpenAI Agents SDK / ADK
- LO2: Develop and deploy Agentic AI and RAG applications for inference on Streamlit Cloud
- LO3: Deploy guardrails and model alignment techniques to ensure safe and aligned outputs in Agentic AI applications

Skills Framework Alignment

Generative AI Application Development and Deployment (ICT-INT-0047-1.1) — Level 1. This TSC covers the development and deployment of generative AI applications, including integration workflows, cloud deployment, and safety guardrails.

Code	Ability / Knowledge statement
A5	Design integration workflows between AI models and systems, such as batch processing, API integration, and multi-model orchestration
A2	Resolve compatibility issues with other software or systems (e.g. GitHub Copilot, CircleCI)
A3	Identify issues in generative AI models by analysing and interpreting error logs
A4	Develop applications that integrate generative AI models via developer tools and cloud service offerings (e.g. Gradio and Streamlit)
A1	Apply guardrails and alignment techniques to safeguard generative AI model outputs
K1	Concepts of software integration
K2	Concepts of integrating multiple generative AI models, including common compatibility and integration issues

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K3	Concepts behind context augmentation techniques, such as knowledge graphs
K5	Approaches to deploy models for inference using cloud platforms
K6	Designs of model pipelines
K4	Techniques to safeguard and align large language model outputs

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on build-and-deploy tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The final assessment is conducted on Day 4 from 4:30 pm (2 hours in total).
- Grading: Competent / Not Yet Competent. An appeal process is available.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Each training day runs 9:30 am – 6:30 pm and delivers exactly 8 instructional hours. The 1-hour lunch break is excluded from instructional time; tea breaks are counted within it. Slide numbers refer to the master trainer slide deck version v2.

Day 1 — Agentic AI Foundations & Your First Agent

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, learning outcomes, ground rules and mandatory AM digital attendance (slide 2)
10:00–11:00	60 min	**LU1 T1 — Introduction to Agentic AI frameworks: the agentic loop, anatomy of an agent, and the CrewAI / AutoGen / OpenAI Agents SDK landscape (slide 18)**
11:00–11:15	15 min	Tea break
11:15–12:30	75 min	LU1 T1 — Framework comparison and selection criteria. Hands-on: Activity 1: Set Up the Agentic AI Development Environment (slide 25)
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Activity 2: Compare CrewAI, AutoGen and the OpenAI Agents SDK (slide 26). Framework decision table and use-case scoring
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	**LU1 T2 — Overview of multi-agent workflows: single-agent vs multi-agent,**

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		specialisation, and the cost of coordination
16:45–18:15	90 min	Hands-on: Activity 3: Build a Single Agent with Tools (OpenAI Agents SDK) (slide 32)
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total facilitation time: 480 minutes (8 hours).

Day 2 — Multi-Agent Workflows & AI-Assisted Development

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory AM digital attendance
9:45–10:45	60 min	**LU1 T3 — Agentic AI workflow design: sequential, router-worker, parallel fan-out and reflection patterns; the six-step design process**
10:45–11:00	15 min	Tea break
11:00–12:30	90 min	LU1 T4 — Build Agentic AI applications with frameworks. Hands-on: Activity 4: Build a Role-Based Crew with CrewAI (slide 34)
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Activity 5: Build a Conversational Team with AutoGen (slide 35)
15:00–15:15	15 min	Tea break
15:15–16:45	90 min	Production patterns: structured output, tool docstrings, failure handling and bounded loops. Hands-on: Activity 6: Orchestrate a Multi-Agent Trip Planner (Agents-as-Tools) (slide 37)
16:45–18:15	90 min	LU1 T5 — Development with AI IDEs (GitHub Copilot, Cursor, Trae, Windsurf). Hands-on: Activity 7: Accelerate Agent Development with an AI IDE (slide 41)
18:15–18:30	15 min	LU1 recap, Q&A and PM digital attendance

Total facilitation time: 480 minutes (8 hours).

Day 3 — RAG, Context Augmentation & Cloud Deployment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory AM digital attendance
9:45–10:45	60 min	**LU2 T1 — Context augmentation: the RAG pipeline, embeddings, vector stores, knowledge graphs and GraphRAG (slide 43)**

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10:45–11:00	15 min	Tea break
11:00–12:30	90 min	Hands-on: Activity 8: Build a RAG Pipeline over Invoice PDFs (slide 48)
12:30–13:30	60 min	Lunch break
13:30–15:00	90 min	Hands-on: Activity 9: Tune Retrieval Quality — Chunking and Top-K (slide 50); Activity 10: Context Augmentation with a Knowledge Graph (slide 51)
15:00–15:15	15 min	Tea break
15:15–16:30	75 min	**LU2 T2 — Approaches to deploy models for inference on cloud platforms; Streamlit Cloud vs Railway. Hands-on: Activity 11: Swap Model Providers — Gemini and OpenRouter (slide 56)**
16:30–18:00	90 min	LU2 T4 — Deploy RAG and Agentic AI apps on Streamlit Cloud. Hands-on: Activity 12: Deploy a RAG App to Streamlit Cloud (slide 57)
18:00–18:30	30 min	LU2 T3 — Analysing and interpreting error logs (concepts). Hands-on: Activity 13: Diagnose Deployment Failures from Error Logs (slide 60). Day 3 recap and PM digital attendance

Total facilitation time: 480 minutes (8 hours).

Day 4 — Guardrails, Model Alignment & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 3 recap and mandatory AM digital attendance
9:45–10:45	60 min	LU2 T3 — Monitoring a deployed multi-agent app: caching, spend controls and token usage. Hands-on: Activity 14: Deploy the Multi-Agent App and Monitor It (slide 61)
10:45–11:00	15 min	Tea break
11:00–12:30	90 min	**LU3 T1 — Safeguarding LLMs with guardrails: threats, three layers of defence, rule-based vs AI-assisted (slide 63). Hands-on: Activity 15: Attack Your Own Agent — Red-Team the Baseline (slide 68)**
12:30–13:30	60 min	Lunch break
13:30–14:45	75 min	Hands-on: Activity 16: Implement Input Guardrails with Tripwires (slide 70); Activity 17: Implement Output

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		Guardrails — PII and Content Filtering (slide 71)
14:45–15:00	15 min	Tea break
15:00–15:30	30 min	**LU3 T2 — Model alignment for safety: RLHF, DPO, Constitutional AI; alignment vs guardrails. Hands-on: Activity 18: Evaluate Model Alignment — Helpful, Harmless, Honest (slide 75)**
15:30–16:15	45 min	LU3 T3 — Deploy guardrails on Agentic AI applications: human oversight, audit logging and the go-live checklist. Hands-on: Activity 19: Add Human Oversight and an Audit Log (slide 78); Activity 20: Deploy the Guardrailed Agentic AI App to Production (slide 80). Course recap and Q&A
16:15–16:30	15 min	**Briefing for Assessment (slide 15), TRAQOM survey and Assessment digital attendance**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — build-and-deploy tasks, 1 hour, open book. PM digital attendance**

Total facilitation time: 360 minutes (6 hours). Final assessment: 120 minutes (2 hours), conducted within the same 8-hour day.

Activity Reference (aligned to Learning Units)

Learning Unit	Outcome	Activities	Slides
LU01: Agentic AI App Development with Agentic AI Frameworks	LO1 · LU1	Activities 1–7	25–41
LU02: Agentic AI and RAG Deployment on Streamlit Cloud	LO2 · LU2	Activities 8–14	48–61
LU03: Model Alignment and Guardrails	LO3 · LU3	Activities 15–20	68–80

Resources and Equipment

- One laptop per learner with Python 3.11+, VS Code and administrator rights to install packages.
- Reliable internet access for model APIs, package installation and cloud deployment.
- API keys: OpenAI, Tavily and OpenRouter (a free tier is sufficient for every activity).

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- Accounts: GitHub and Streamlit Community Cloud (both free) for the deployment activities.
- Course repository: https://github.com/tertiarycourses/TGS-2025059028-Build-and-Deploy-Agentic-AI-Apps-with-CrewAI-Autogen-ADK-and-Streamlit
- Master trainer slide deck, Learner Guide and lab files distributed via lms-tms.tertiaryinfotech.com.

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